

Week 3

9.20 星期四

$$11.18) \quad \text{令 } z = iy, \quad \text{则 } \lim_{z \rightarrow 0} z \cdot \sinh \frac{1}{z} = \lim_{y \rightarrow 0} iy \cdot \frac{e^{\frac{1}{iy}} - e^{-\frac{1}{iy}}}{2i} \\ = \lim_{y \rightarrow 0} \frac{y}{2} \cdot (e^{\frac{1}{y}} - e^{-\frac{1}{y}})$$

$$\text{当 } y \rightarrow 0 \text{ 时, } y \rightarrow 0^+ \quad \text{s.t.} \quad \lim_{y \rightarrow 0^+} e^{\frac{1}{y}} = +\infty$$

$$y \rightarrow 0^- \quad \text{s.t.} \quad \lim_{y \rightarrow 0^-} e^{\frac{1}{y}} = 0$$

$$\therefore \lim_{y \rightarrow 0} \frac{y}{2} (e^{\frac{1}{y}} - e^{-\frac{1}{y}}) \rightarrow \infty$$

$\therefore$ 极限不存在

$$12) \quad \text{令 } z = 1+x \quad \text{则 } x \rightarrow 0$$

$$\text{则 } \lim_{z \rightarrow 1} \frac{z \cdot e^{\frac{1}{z-1}}}{e^{\frac{1}{z-1}} - 1} = \lim_{x \rightarrow 0} \frac{(1+x) \cdot e^{\frac{1}{x}}}{e^{\frac{1}{x}} - 1}$$

同 11) 中理由, $\lim_{x \rightarrow 0} e^{\frac{1}{x}}$ 不存在

$\therefore$ 极限不存在

$$13.10) \quad \sinh z = 2 \Rightarrow \sinh z = \frac{1}{2i} \cdot (e^{iz} - e^{-iz}) = 2$$

$$\Rightarrow e^{iz} = (2 \pm \sqrt{3})i$$

$$\therefore z = -i \cdot \frac{1}{i} \cdot \ln(2 \pm \sqrt{3})i$$

$$= \left(\frac{\pi}{2} + 2k\pi\right) - i \ln(2 \pm \sqrt{3})$$

$$\text{or } z = \text{Arcsinh } 2 = -i \ln(2 \pm \sqrt{3})i$$

$$= \left(\frac{\pi}{2} + 2k\pi\right) - i \ln(2 \pm \sqrt{3})$$

$$14.12) \quad \frac{1}{\sinh z - 2} \Rightarrow \sinh z \neq 2$$

$$\Rightarrow z \neq \left(\frac{\pi}{2} + 2k\pi\right) - i \ln(2 \pm \sqrt{3}) \text{ 为解析区域}$$

$$\left(\frac{1}{\sinh z - 2}\right)' = \frac{-\cosh z}{(\sinh z - 2)^2}$$

$$13) \quad z \cdot e^{\frac{1}{z-1}} \Rightarrow z \neq 1 \text{ 为解析区域}$$

$$(z \cdot e^{\frac{1}{z-1}})' = e^{\frac{1}{z-1}} + z \left(-\frac{1}{(z-1)^2}\right) \cdot e^{\frac{1}{z-1}} = e^{\frac{1}{z-1}} \cdot \left(1 - \frac{z}{(z-1)^2}\right)$$

$$16. \cos z = \cos x \cdot \cosh y - i \sin x \cdot \sinh y \in \mathbb{R}$$

$$\Rightarrow \sin x \cdot \sinh y = 0 \Rightarrow \begin{cases} x = k\pi \\ y = 0 \end{cases} \text{ or}$$

$\therefore$ 实轴, 以及直线族 $\operatorname{Re} z = k\pi \ (k \in \mathbb{Z}) \perp, \cos z \in \mathbb{R}$

$$17. (z) \quad 1^{\sqrt{z}} = e^{\sqrt{z} \ln 1} = e^{\sqrt{z} (\ln 1 + i \cdot 2k\pi)} = e^{i 2\sqrt{z} k\pi}$$

$$(-z)^{\sqrt{z}} = e^{\sqrt{z} \ln(-z)} = e^{\sqrt{z} (\ln z + (2k+1)i\pi)}$$

$$z^i = e^{i \ln z} = e^{i (\ln z + i 2k\pi)} = e^{i \ln z - 2k\pi}$$

$$(3-4i)^{1+i} = e^{(1+i) \ln(3-4i)} = e^{(1+i) [\ln 5 + i (\arg(3-4i) + 2k\pi)]}$$

$$= e^{[\ln 5 - \operatorname{arctg}(-\frac{4}{3}) - 2k\pi] + i [\ln 5 + \operatorname{arctg}(-\frac{4}{3}) + 2k\pi]}$$